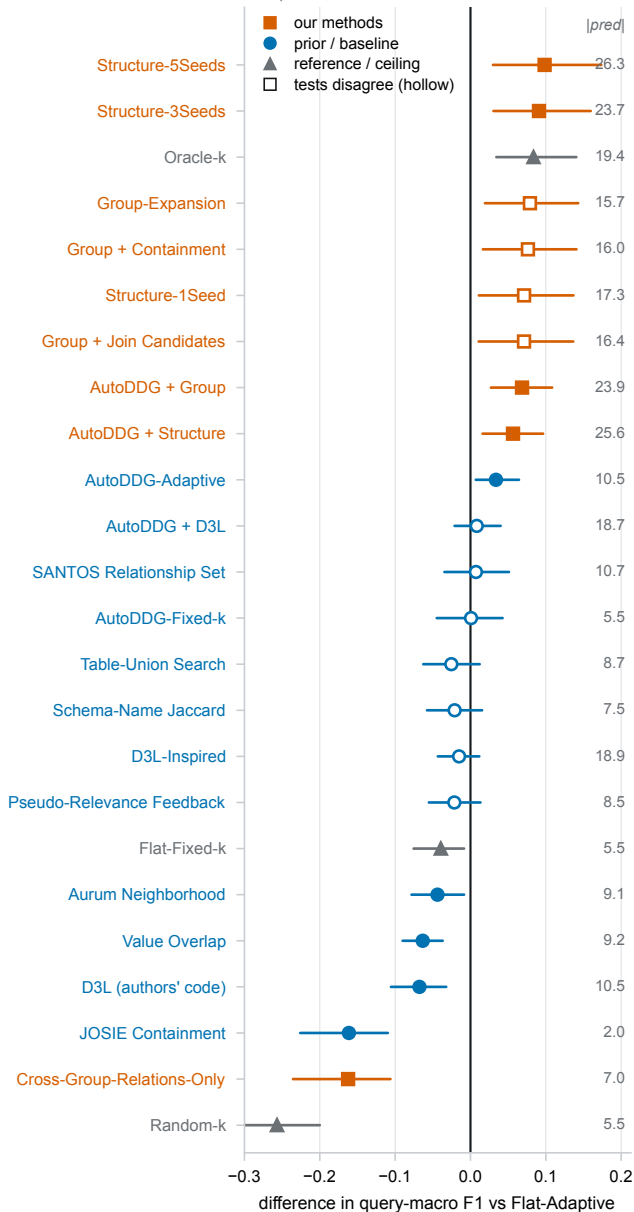


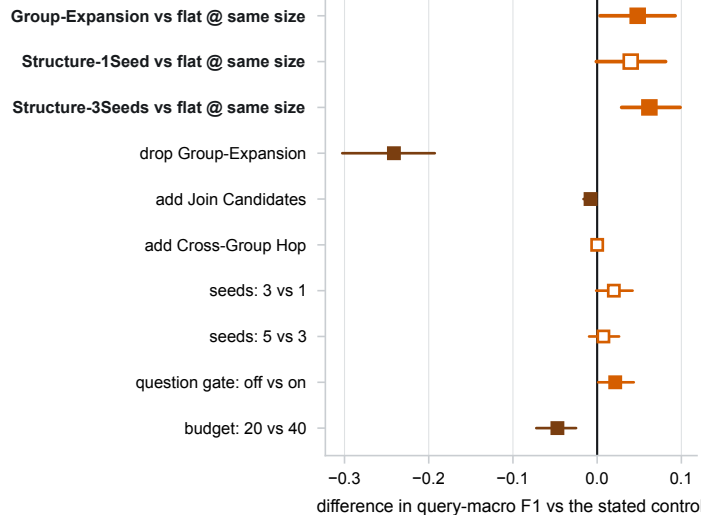
A Comprehensive retrieval comparison

245 queries, 2 benchmarks, reference arm F1 = 0.270



B Component ablation and controls

each contrast is paired and uses its own stated control



C Query-level effect heterogeneity

